Emergence of Born Statistics from the Equidistribution of the Observer–System Beat — A Conjecture on the Localized-Kernel Model

Noriaki Kihara (Derived from peer-review dialogue notes. **This is neither a proof paper nor an observation paper, but a conjecture (prediction) paper.** It is not a claim of established physics, but the presentation of a candidate mechanism, made explicit, together with its testable consequences.)

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Revision note (v0.3): (1) Removed the §7 clause "the gradient energy $\sum \nu^2$ becomes exactly finite," which suggested an energy divergence (the falsifiable prediction itself — the $O(1/N)$ reproduction correction — is unchanged). (2) Made explicit at the end of §2 that the **match of frequency is self-evident** (the localized wave is sculpted in **shape** by odd harmonics but its fundamental frequency $\nu = 1$ is unchanged = scale invariance), sharpening the division of labor between Point 1 (self-evident frequency match + non-trivial shape reproduction) and Point 2 (non-trivial fluctuation = beat). (3) Echoed in the §9 conclusion the role separation "**Weyl equidistribution supplies only a uniform measure; the $|\psi|^2$ bias comes from the reproducing kernel + the bridge**" (previously only in §5(D)). Claims, conclusions, and equations are unchanged.

Revision note (v0.2) (reflecting the second review, Claude Code): (1) both input cases derived explicitly — localized input and spread input $\cos\varphi$ come from the same reproducing kernel, the output difference being only the input difference (§3, §5(B), confirmed by actual convolution to machine precision); (2) §3(I) role separation rewritten — equidistribution (Weyl) supplies a uniform measure (no bias), the $|\psi|^2$ shape is the deterministic reproducing-kernel read, intensity→click is the bridge (§3 II); the conflation "Weyl alone makes the squared bias" is removed; (3) falsification battery (§5 E) — spread input not peaked, localized input peaked, Weyl alone flat, both cases one identity — all four pass; (4) §5(B) upgraded from a restatement of [2] to a self-contained verification by actual convolution (tautology removed), with the band condition (N-dependence) shown; (5) the confirmed DOI of [2] added; (6) the "proven core" made conditional on the equidistribution assumption (§8 O1), and the discrete-clock dependency (B1 upstream of B2/B3) made explicit (§2, §8 O2); (7) the mononym "Lynnx" in [7] confirmed as listed on arXiv; (8) the status of the §5(D) bridge Monte-Carlo made explicit — it is a consistency demonstration that *assumes* the bridge (§3 II), not a derivation/verification of it. **Claims, conclusions, and equations are unchanged** (rigor and honesty only).

Abstract

The preceding localized-kernel paper [2] showed, from the **reproducing-kernel property** of the constant-amplitude odd-harmonic sum (isolated peak wave S_N) on the half-wavelength phase interval $\varphi \in [-\pi/2, \pi/2]$, that the **form** of the observed distribution coincides exactly with the base wave's $|\psi_{\mathrm{base}}|^2$. It left, however, "the existence of randomness" — why a deterministically computed interference-intensity profile localizes to a single point each trial and appears as the frequency $|\psi_{\mathrm{base}}|^2$ only upon repetition — as an unsolved postulate.

This paper presents a candidate mechanism for this **origin of randomness** as a **conjecture**. The central observation is that the scan variable φ_0 of the localized-kernel paper is **not a free variable but the phase of the observer**. The observer is itself a physical object with a finite frequency ψ, and there is no absolute phase reference outside the system (the subjective-space premise). Hence the measurement sequence scans the **relative phase** (beat/heterodyne) $(\nu - \psi)t$ of the system frequency ν and the observer frequency ψ. When the relative-phase ratio is irrational, this relative phase **equidistributes** over the interval (Weyl's equidistribution theorem).

The logic has two stages. **(I) Conditionally proven core**: by the reproducing-kernel property [2], the squared overlap at each observer phase φ_0 is exactly $(\pi/2)^2|\psi_{\mathrm{base}}(\varphi_0)|^2$, and **if** the beat phase equidistributes (irrational ratio, §8 O1), the long-time **position-resolved intensity profile** coincides, after normalization, with $|\psi_{\mathrm{base}}|^2$. What is guaranteed here is the **shape of the deterministically swept intensity profile**, not a single-shot probability. This consequence follows, granting the equidistribution assumption (O1), from the reproducing-kernel property (proven) and Weyl's theorem (proven) with no further assumption. **(II) Conjecture (bridge)**: we conjecture the transition by which this intensity profile appears, via threshold detection, as the **frequency of single clicks** $\propto |\psi_{\mathrm{base}}|^2$. (II) is the part this paper does **not** derive; why a single point stands up each shot (discreteness) is left unsolved here.

This conjecture is not an isolated idea but sits at the **intersection** of two established lineages: (a) Khrennikov's prequantum classical statistical field theory (PCSFT) and the emergence of the Born rule via threshold detection [3–6] (the physical picture), and (b) the **method of arbitrary functions** [7–10], begun by Poincaré and applied to the quantum case by Feintzeig et al. (the mathematical mechanism = our equidistribution). The contribution specific to this paper is to physically identify the "fast equidistributing random variable" that both lineages place abstractly as the observer–system **heterodyne beat**, and to fix the shape of the intensity profile **exactly** by the reproducing-kernel property. We further present a falsifiable prediction: that the reproduction error $O(1/N)$ at finite N (discrete lattice) gives a **measurable deviation from the Born rule**.

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1. Background and the place of this paper

The localized-kernel paper [2] derived the "form" of the Born rule $P \propto |\psi|^2$ squared structure from the reproducing-kernel property, narrowing the remaining postulates to two: (a) the **squaring rule** of amplitude → probability (exponent 2), (c) the **existence of randomness**. A **measurement model** (b) representing measurement by the convolution of the localized kernel also remains a premise. What [2] rigorously established is: for a band-limited base wave $\psi_{\text{base}} = \sum_{|k|\le N} c_k e^{ik\varphi}$, the convolution with the shifted localized kernel S_N satisfies

$$(S_N * \psi_{\text{base}})(\varphi_0) = \frac{\pi}{2}\,\psi_{\text{base}}(\varphi_0) \quad\Longrightarrow\quad \big|(S_N * \psi_{\text{base}})(\varphi_0)\big|^2 = \frac{\pi^2}{4}\,|\psi_{\text{base}}(\varphi_0)|^2$$

exactly over the whole interval and for all N (a genuine modulus $Z\bar{Z}$ in the complex case).

This paper takes up (c). It is the deepest of the three postulates, the measurement problem itself, unsolved even in standard theory. We do **not claim to have solved it**; we formulate a candidate mechanism as a **conjecture**, show that part of it (the reproduction of the intensity profile) follows from the proven core, and explicitly carve out the rest (discreteness, the probability interpretation) as unsolved.

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2. Sharpening the problem: who decides φ_0?

In [2], φ_0 is plotted over the whole interval as a free variable. But **one observation occurs at only one** φ_0. The deterministic computation gives the function $|\psi_{\text{base}}(\cdot)|^2$, yet one measurement returns a single realized value. This transition "**function → a single realization per trial**" is the entire remaining mystery. We decompose it into three logically distinct bridges:

- **B1 (discreteness)**: why only one result stands up each time (the whole profile does not appear at once).
- **B2 (which one)**: granting discreteness, why which φ_0 is unpredictable from the state.
- **B3 (ensemble = intensity)**: why the trial frequency matches the intensity profile.

The weight $\propto |\text{overlap}|^2$ (a golden-rule-like coupling rate) is the least non-trivial of the three. Positional fluctuation is the manifestation of B1 · B2, not a mystery of the **form** of $|\psi|^2$ — the form is already settled in [2] as the intensity. This paper's conjecture mainly answers B2 and B3, and leaves B1 unsolved.

However, the three bridges are **not fully independent**. The equidistribution mechanism below (§3–§4) becomes non-trivial only on a **discrete sequence of measurement events** — in continuous time the sweep of $\varphi_0 = (\nu - \psi)t$ becomes uniform without irrationality and the rational counter-example disappears — so B1 (discreteness) is logically **upstream** of the B2 · B3 mechanism. That is, the equidistribution argument tacitly presupposes a discrete clock. This dependency is made explicit in §8 O2.

Note that the appearance of the observation at the same **frequency** as the baseline $\nu = 1$ probability wave is itself self-evident: although the localized wave is sculpted in **shape** by odd harmonics, the fundamental frequency $\nu = 1$ of the composite wave is unchanged (since $\gcd(1, 3, \ldots, N) = 1$, the fundamental period is set by the lowest mode $\nu = 1$; this is merely an observation under the scale-invariant relative normalization, made explicit in the odd-harmonic localization paper). Hence what this paper asks is not the frequency match, but **(a) the mechanism by which the observed waveform is faithfully reproduced as** $|\psi_{\mathrm{base}}|^2$ (the reproducing-kernel property, [2]; non-trivial because a general kernel would distort it) and **(b) the mechanism by which observation fluctuates across trials** (the beat). Point 1 (frequency match = self-evident, scale invariance / shape reproduction = reproducing kernel) and Point 2 (fluctuation = beat) belong to different layers.

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3. Conjecture: equidistribution of φ_0 by the observer–system beat

Premise (the only physics added). The observer is a physical object with a finite frequency ψ. A perfect absolute phase reference (an absolute clock outside the system) does not exist — this is a necessity of the subjective-space premise shared by [2] and this paper (only differences are physical; absolute phase is unobservable), not a new assumption. If the observer were a perfect reference, φ_0 would be fixed and the fluctuation would vanish, but the premise forbids such a device.

Beat. The product of the system $\cos(\nu t)$ and the observer $\cos(\psi t)$ is

$$\cos(\nu t)\cos(\psi t) = \tfrac{1}{2}\left[\cos((\nu - \psi)t) + \cos((\nu + \psi)t)\right]$$

so the difference frequency $\nu - \psi$ gives a slow envelope (the fringe period). Because the observer phase advances at each measurement timing, the effective scan phase is $\varphi_0 = (\nu - \psi)t$. If the observer has the same frequency as the system ($\psi = \nu$), φ_0 is fixed and the same phase is read each time — no fringes, a single value. If the observer differs ($\psi \neq \nu$), φ_0 is scanned and repetition sweeps out the distribution.

Determinism of the single shot. Importantly, the single convolution readout at each φ_0 is, by [2], **deterministic**: $(S_N * \psi_{\mathrm{base}})(\varphi_0) = \frac{\pi}{2}\psi_{\mathrm{base}}(\varphi_0)$ is sharp and unambiguous. This is consistent with the fact that the observed peak wave is actually sharply localized (it does not scatter randomly). The ambiguity resides not in the single shot but in the **distribution of φ_0 over repetitions**.

Conjecture (main). When measurements occur at discrete times t_n (by the observer clock) and the relative-phase ratio $r = (\nu - \psi)\,\Delta t/(2\pi)$ is irrational, $\varphi_0 = (\nu - \psi)t_n$ **equidistributes** over $[-\pi/2, \pi/2]$ (Weyl's equidistribution theorem for an irrational rotation). The reliance on a discrete measurement sequence (in continuous time irrationality is unnecessary and the counter-example disappears) corresponds to the dependency of §2. Hence in the long run each φ_0 is uniformly sampled, and the long-time average of the position-resolved squared readout is

$$\boxed{\lim_{T\to\infty}\frac{1}{T}\int_0^T \big|(S_N*\psi_{\mathrm{base}})((\nu-\psi)t)\big|^2\,\delta_{\varphi_0}\,dt \;\propto\; |\psi_{\mathrm{base}}(\varphi_0)|^2}$$

i.e. the **position-resolved intensity profile** coincides, after normalization, with $|\psi_{\mathrm{base}}|^2$. This equality follows, **granting the equidistribution assumption (§8 O1), with no further assumption**, from the reproducing-kernel property (the exact result of §1) and Weyl equidistribution (a classical theorem) (confirmed numerically in §5). What is guaranteed is the **shape** of the intensity profile, not a single-shot probability (the bridge is below).

Role separation (removing the conflation). Here we rigorously separate three contributions. **Equidistribution (Weyl) only supplies the uniform measure $d\varphi_0$; by itself it makes no squared bias** — if the observer phase φ_0 is swept uniformly, all positions are equally frequent (constant), which is not the $|\psi|^2$ bias. The **source of the per-position weight (shape) $|\psi_{\mathrm{base}}(\varphi_0)|^2$ is the deterministic reproducing-kernel read at each** φ_0 ([2]: $(S_N*\psi_{\mathrm{base}})(\varphi_0)=\frac{\pi}{2}\psi_{\mathrm{base}}(\varphi_0)$, reproducing the input ψ_{base} as is), not equidistribution. And what turns the intensity profile into the frequency of single clicks is the **threshold-detection bridge** (§3 (II)). Schematically:

$$\underbrace{\text{frequency of }|\psi_{\mathrm{base}}(\varphi_0)|^2}_{\text{Born output}}=\underbrace{d\varphi_0\ \text{(uniform)}}_{\text{Weyl equidistribution}}\quad\times\quad\underbrace{|\psi_{\mathrm{base}}(\varphi_0)|^2}_{\text{deterministic kernel read (shape)}}\quad\times\ \underbrace{\text{(intensity}\to\text{click)}}_{\text{bridge, §3 (II)}}.$$

The **proven core** asserts only the middle factor (the intensity-profile shape = $|\psi_{\mathrm{base}}|^2$), with Weyl supplying the left uniform measure. The right bridge is conjecture. We do not write as if Weyl alone makes the bias (in §5 D we also confirm numerically that the histogram of equidistributed φ_0 is **flat**, and $|\psi|^2$ appears only through intensity weighting).

Exact derivation of both cases (same mechanism, input-dependent only). Since the reproducing-kernel identity $(S_N*\psi_{\mathrm{base}})(\varphi_0)=\frac{\pi}{2}\psi_{\mathrm{base}}(\varphi_0)$ holds independently of the input ψ_{base}, for two representative inputs:

- **Case 1 (localized)**: for $\psi^{\mathrm{loc}}_{\mathrm{base}}=\sum_{|k|\le N}c_k e^{ik\varphi}$ (a multimode superposition localized at the center; the most localized example is S_N itself),

$$\big|(S_N*\psi^{\mathrm{loc}}_{\mathrm{base}})(\varphi_0)\big|^2=\tfrac{\pi^2}{4}\,|\psi^{\mathrm{loc}}_{\mathrm{base}}(\varphi_0)|^2\ \text{(peaked output)}.$$

- **Case 2 (spread)**: for $\psi^{\cos}_{\mathrm{base}}=\cos\varphi$ (a spread single mode),

$$\big|(S_N*\cos)(\varphi_0)\big|^2=\tfrac{\pi^2}{4}\cos^2\varphi_0\ \text{(spread output)}.$$

Both come from the **same reproducing kernel**, and the output difference is only the input ψ_{base} difference. **A spread input (cos) gives a spread output** ($\cos^2$); **a localized input gives a peaked output** — since the reproducing kernel reproduces the input without distortion, there is no breakdown beyond the input dependence. Case 2 is numerically supported by Fig. 1(a) of [2] (the cos base lies on

$\cos^2$ for all N), and is included as an explicit test in this paper's verification code (§5). This **agreement of both cases** is the strongest evidence that the mechanism is consistent with input dependence only (clearing all falsification conditions) (§5 E).

Conjecture (bridge, unproven). The above intensity profile appears, via threshold detection, as the **frequency of single clicks** $\propto |\psi_{\rm base}|^2$. That is, B3 (frequency = intensity) is reduced to the intensity profile, and B1 (single-shot) is delegated to threshold detection. This bridge is the part this paper does **not** derive, connecting to the mechanism carried by lineage A (Khrennikov) of §6.

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4. Why determinism and probability coexist: the method of arbitrary functions

The mathematical body of the mechanism of §3 is the **method of arbitrary functions**. This is the classical-probability tradition in which a fast-varying phase (or dynamical parameter) and **any smooth distribution** over it converge, in a certain limit, to a **universal output distribution** independent of the input details [8–10]. Poincaré analyzed the rotation of a roulette wheel and showed that any smooth distribution of the initial angular velocity gives nearly equal probability to each pocket at high speed [8]. Diaconis et al. gave the same kind of result for coin tossing [17]. The point of the method is to **reconcile non-trivial probabilities (neither 0 nor 1) with a deeper layer of determinism**.

Our setup is the quantum version of this. Just as fast rotation equidistributes over the **alternating red-black arcs** of Poincaré's wheel (a periodic binary structure on the circle), so our measurement reads a **0/180 binary sign** (the sign of the real-axis projection), and the high-frequency beat equidistributes the relative phase — the same mathematical object: equidistribution of a fast phase over a periodic structure. Feintzeig et al. [7] showed that treating a dynamical parameter as a random variable with an initial distribution yields the Born rule as **universal limiting behavior**, for any initial distribution, in the joint long-time and classical limit. This paper physically identifies that "equidistributing dynamical parameter" as the observer–system **relative phase**.

Therefore randomness is intrinsic neither to the state nor to the observer alone. It emerges from the **relative phase of two deterministic oscillators** equidistributing over the measurement sequence. This breaks the circularity: $|\psi|^2$ is assumed nowhere; the fluctuation is produced from the independent fact of the boundary of observability (the absence of absolute phase).

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5. Numerical corroboration (the testable core)

We confirm numerically the **(I) proven core** of §3. Base wave $\psi_{\rm base} = \cos\varphi + 0.5\cos 3\varphi - 0.3\cos 5\varphi$, $N = 9$.

(A) Weyl equidistribution of the beat phase. Taking the relative-phase ratio irrational (golden ratio $(\sqrt{5}-1)/2$), the star-discrepancy from the uniform distribution when φ_0 is sampled by the irrational rotation:

beats N_{beats}	10^2	10^3	10^4	10^5
star-discrepancy	1.4×10^{-2}	1.3×10^{-3}	2.6×10^{-4}	3.0×10^{-5}

Converges to 0 as N_{beats} grows (equidistribution). By contrast, for **rational ratios** $(1/2, 1/3, 1/4)$ the discrepancy is stuck at 0.25–0.50 and does not equidistribute (the fringe is biased; counter-example). That is, "the observer has a generic (irrational-ratio) frequency relative to the system" is the condition for equidistribution.

(B) Actual convolution of the reproducing kernel (both cases). In v0.2 we **actually integrate** $(S_N * \psi_{\text{base}})(\varphi_0) = \int_{-\pi/2}^{\pi/2} S_N(\varphi_0 - \varphi)\psi_{\text{base}}(\varphi)\, d\varphi$ and confirm agreement with $\frac{\pi}{2}\psi_{\text{base}}(\varphi_0)$ self-containedly in this paper's code (not merely restating [2]). Machine-precision agreement for all three representative inputs:

input ψ_{base}	$\max \lvert (S_N * \psi) - \frac{\pi}{2}\psi \rvert$	$\max \big\lvert \lvert \cdot \rvert^2_{\text{norm}} - \lvert\psi\rvert^2_{\text{norm}} \big\rvert$
Case 1 localized $\psi^{\text{loc}} = S_N$	1.8×10^{-15}	8.9×10^{-16}
Case 1' multimode $\cos + 0.5\cos 3 - 0.3\cos 5$	4.4×10^{-16}	4.4×10^{-16}
Case 2 spread $\cos\varphi$	4.4×10^{-16}	3.3×10^{-16}

From the **same reproducing kernel**, the output difference is only the input difference: localized input $\to$ peaked output, spread input $\to$ spread output (numerical confirmation of the both-case derivation of §3).

Band condition (N-dependence). The reproducing kernel S_N reproduces only modes $|k| \leq N$. Verifying by varying N (each cell is $\max|(S_N * \psi) - \frac{\pi}{2}\psi|$):

input	$N = 1$	$N = 3$	$N = 5$	$N = 9$	$N = 31$
$\cos\varphi$ (mode 1)	2e-16	2e-16	4e-16	4e-16	2e-16
multimode (1,3,5)	1.2	0.47	4e-16	4e-16	7e-16

cos is **exact for all** N; the multimode (highest degree 5) is **exact for** $N \geq 5$ **and dropped for** $N < 5$. This is the manifestation of premise (ii) "ignore unobservable phase differences" = truncation exact once it covers the band. Note (B) is not the previous "identity by substituting $(\pi/2)\psi$" but a **self-contained independent verification by actually convolving** S_N (the previous tautology is removed).

(C) Single complex mode. For $\psi = e^{ik\varphi}$ ($k = 1, 3, 5$) the position-dependent spread of the squared readout is $\sim 3 \times 10^{-15}$ (=0, uniform $= |e^{ik\varphi}|^2 = 1$). A sharp single-frequency mode carries no position information and is fully delocalized — a manifestation of $\Delta\nu \cdot \Delta\varphi \gtrsim 1$.

(D) Numerical separation of roles (Weyl alone makes no bias). We directly check the role separation of §3. The **histogram itself** of φ_0 equidistributed by the irrational rotation (2×10^6 points) is **flat**, with a maximum deviation from the uniform density $1/\pi$ of 1.0×10^{-5} — **equidistribution makes no bias**. On the other hand, when each φ_0 visit is accepted as a click with probability proportional to the intensity $|\psi_{\text{base}}(\varphi_0)|^2$ (a Monte-Carlo model of the threshold-detection bridge), the histogram of accepted clicks matches $|\psi_{\text{base}}|^2$ (Case 2 cos: deviation 2.7×10^{-3}; Case 1' multimode: 6.7×10^{-3}; both MC statistical error). That is, the $|\psi|^2$ bias comes from the **deterministic kernel read + the bridge**, not from equidistribution.

Important (status statement). This D2 Monte-Carlo is a **consistency demonstration that assumes the bridge (§3 II) from the outset** ("make clicks with probability proportional to the intensity $|\psi_{\text{base}}(\varphi_0)|^2$"); it is **neither a derivation nor a verification of the bridge**. If you accept in proportion to intensity, getting $|\psi|^2$ out is automatic; what is shown here is only the consistency that "assuming the bridge closes the whole thing without contradiction (uniform measure × intensity profile × bridge = $|\psi|^2$ frequency)." Hence the status separation of this paper does not collapse: the **core** = Weyl (proven) × reproducing kernel (proven) = uniform measure × the **shape** $|\psi_{\text{base}}|^2$ of the intensity profile (backed by D1 and (B)); the **bridge** = intensity → single-click probability (**conjecture**, in D2 merely assumed; delegated to threshold detection = lineage A). D2 does **not** change the status that the intensity is not yet a probability (the intensity profile is a classical time-averaged interference fringe).

(E) Falsification battery (consistency conditions). We check that the mechanism does not break with "input dependence only," under conditions that, if violated, would make the theory wrong:

1. **Does a spread input give a peaked output?** The output peakedness (max/mean) of the cos input is 1.96 (low = stays spread). **OK** (no breakdown).
2. **Does a localized input over-spread?** The output peakedness of the S_N input is 9.80 (high = stays peaked). **OK**.
3. **Does Weyl alone make a squared bias?** Confirmed flat in (D1). **OK**.
4. **Can Case 2 be false independently of Case 1?** Both are special cases of the one identity $(S_N * \psi) = \frac{\pi}{2}\psi$ and cannot be refuted independently (on the same bridge). **OK**.

All four conditions pass. This is a confirmation that "both cases come out correctly with the same mechanism and input dependence only," and is **positive (offensive) evidence for the consistency of the conjecture**.

The product of (A) (Weyl = uniform measure) and (B) (reproducing kernel = deterministic read of the shape) constitutes the boxed equality of §3 — the **shape** of the position-resolved intensity profile $= |\psi_{\text{base}}|^2$. This is the paper's **(conditional on the equidistribution assumption O1) proven core**. (D)(E) corroborate the role separation and the consistency. The remaining bridge (intensity → single-shot probability) is a claim of a kind not confirmable numerically, treated in §6 · §8.

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6. Relation to prior work

This conjecture sits at the intersection of two established lineages.

6.1 Lineage A: Khrennikov's PCSFT + threshold detection (the physical picture)

In Khrennikov's prequantum classical statistical field theory, a quantum system is a symbolic representation of a classical random field; Schrödinger dynamics is a special form of the linear dynamics of classical fields; measurement is the interaction of the classical field with a detector; and "wave-packet collapse" = discontinuity has the **trivial origin of the use of a threshold detector** [6]. The detection threshold is set by the average energy of the prequantum signal, and the Born rule is reproduced as the measurement of a classical signal by a properly calibrated threshold detector [3,4]. The presence of a background field (vacuum fluctuation) is a key element [4].

Agreement with this paper: **discrete detection comes from the detector, not the state** (B1); **the measuring device has its own physics**. Difference: Khrennikov's source of randomness is a **stochastic prequantum field** (L^2-valued Wiener process / Gaussian field) [5], and the Born rule is **approximate** and predicts violations. Our source of randomness is the **deterministic beat** (equidistribution of the relative phase), and by the reproducing-kernel property the form is **exact**. The "bridge" of our conjecture (intensity $\to$ single click) corresponds exactly to the mechanism carried by Khrennikov-type threshold detection. That is, this paper cites lineage A as a **supplier of the mechanism**.

6.2 Lineage B: the method of arbitrary functions (the mathematical mechanism)

As in §4, our equidistribution argument is the method of arbitrary functions itself, begun by Poincaré [8], Hopf [9], Engel [10], with the quantum application by Feintzeig et al. [7]. The contribution specific to this paper is to physically identify the "dynamical parameter taken as a random variable," which they place abstractly, as the observer–system **heterodyne relative phase** (extending the line by which [2] grounded it in the Ramsey/heterodyne metrology of [15]), and to fix the shape of the intensity profile exactly by the reproducing-kernel property.

6.3 Adjacent lineages and typicality

On the foundational side, 't Hooft's deterministic (cellular-automaton) quantum mechanics [12] stands alongside as a discrete substrate, and Buniy–Hsu–Zee "Discreteness and the origin of probability in quantum mechanics" [11] belongs to the discretization-first stance. Also Zurek's envariance [13] and the typicality of Everett-type accounts belong to the **same "symmetry/typicality $\to$ weight" family** as the method of arbitrary functions. Our equidistribution (B3), the equal amplitude (reproducing kernel) of [2], and the equiprobable branches of envariance can be seen as different faces of one family.

6.4 Summary of placement

The contribution of this paper is clear as an intersection: **Khrennikov-type measurement-induced discreteness (lineage A) + the universal equidistribution weight of the method of arbitrary functions (lineage B) + the exactness of the form by the reproducing kernel ([2]) + the heterodyne identification of the equidistributing fast variable (this paper)**. This combination is not found in the existing literature.

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7. A testable prediction: the finite-N correction = a Born-rule violation

It is suggestive that lineages A and B both yield the Born rule only in a **limit/approximation**: Khrennikov approximately (and predicts violations) [3], the method of arbitrary functions in the joint long-time and classical limit [7]. Our conjecture too is appropriately stated as the universal/typical behavior of the Born rule in the **equidistribution limit** (irrational beat, large N), and **has a correction at finite values**.

Concretely, replacing the continuous-interval kernel S_N of [2] by an **odd-harmonic kernel on a finite lattice** (a kernel hard-cut at the Nyquist of discrete spacetime), the reproduction is no longer exact, carrying an $O(1/N)$ correction. Here $N = R/\ell$ is the scale ratio (the system localization width ℓ and the interval R). This correction is observable as a **position-dependent deviation from the Born rule**, of size $\sim 1/N = \ell/R$, predicted to be largest at the band edge (near the high-order modes). This is in the same experimental scope as lineage A (the violation Khrennikov predicts), making the conjecture **falsifiable**.

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8. Open problems

We make explicit what this paper does **not** derive.

- **O1 (the equidistribution assumption)**: for the position-resolved intensity to match $|\psi|^2$, the relative phase must equidistribute with an irrational ratio (the condition confirmed in §5). This is natural but an assumption, and B3 is not fully eliminated but **relocated** to "equidistribution of the observer phase." Still, this is far weaker than directly assuming $|\psi|^2$.
- **O2 (B1, single-shot)**: the beat gives the **distribution** of φ_0 but does not give why a **single** click stands up per trial. This is a **separate mechanism** outside this paper's scope, also a part carried by lineage A (threshold detection). Note that the equidistribution argument of §3–§4 (irrational rotation → Weyl) presupposes a **discrete sequence of measurement events** (in continuous

time the sweep is uniform without irrationality, and the rational counter-example disappears). **This paper leaves this discrete measurement sequence as an assumption and does not enter into its origin (the discretization mechanism).** Hence the discrete clock of B1 is also a **premise** of the B2 · B3 mechanism, and O2 is not fully independent of the other open problems. Equidistribution (O1) and single-shot-ness (O2) share the common upstream of "the existence of discrete measurement events."

- **O3 (the power of the square)**: the **exponent 2** that turns amplitude into probability should come from the area measure (Robertson-type) of the complex structure (real 2D), and does not come from the beat. It belongs to the complex-modulus argument $Z\bar{Z} = \mathrm{Re}^2 + \mathrm{Im}^2$ of [2].
- **O4 (scale attribution)**: in a map of type $\sum \nu_n^2 = \nu^2$, one must fix which of ν and the internal ν_n is the higher scale. The single frequency ν seen by the observer and the direction of screening of the internal fine phase depend on this attribution.

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9. Conclusion

This paper presented, as a **conjecture**, a candidate mechanism for the **origin of randomness** left as a postulate by the localized-kernel paper [2]. The central observation is that the scan phase φ_0 is the observer's phase; that, since the observer is a physical object with a finite frequency, no absolute reference exists (the subjective-space premise); hence the measurement sequence scans the observer–system relative phase (the beat), which equidistributes with an irrational ratio (Weyl).

The logic is two-stage. **(Conditional on O1) proven core**: from the reproducing-kernel property ([2]) and Weyl equidistribution, the **position-resolved intensity profile** swept by the beat coincides with $|\psi_{\text{base}}|^2$ (granting equidistribution O1, with no further assumption; confirmed numerically in §5). **Conjectured bridge**: this intensity appears, via threshold detection, as the frequency of single clicks $\propto |\psi_{\text{base}}|^2$ (this paper does not derive it; it connects to lineage A). Here Weyl equidistribution supplies only a **uniform measure (the locus of the fluctuation)**; the $|\psi_{\text{base}}|^2$ bias comes from the **deterministic reproducing-kernel read + the bridge** (§5 D). That the observation coincides with the baseline frequency $\nu = 1$ is itself self-evident, since the harmonics change only the waveform and not the fundamental frequency (scale invariance); what is non-trivial is the **faithful reproduction of the waveform (reproducing kernel) and the origin of the fluctuation (beat)**.

This conjecture sits at the intersection of two established lineages — Khrennikov's PCSFT + threshold detection (the physical picture) and the method of arbitrary functions (the mathematical mechanism) — and its specific contribution is to physically identify the equidistributing fast variable as the observer–system heterodyne beat and to make the form exact by the reproducing-kernel property. Furthermore, the $O(1/N) \sim \ell/R$ correction at finite N (discrete lattice) gives a falsifiable prediction of a measurable deviation from the Born rule.

What remains is B1 (single-shot), the exponent 2, and the scale attribution, which this paper honestly carves out as unsolved. We have not "derived" the Born rule; the scope of this paper is to present, in a testable form, a **candidate that relocates its statistical origin to two independently correct ingredients — the finiteness of the observer and the equidistribution of the relative phase**.

Appendix A: Sketch of the minimal verification

Base wave $\psi_{\text{base}} = \cos\varphi + 0.5\cos 3\varphi - 0.3\cos 5\varphi$, $N = 9$.

1. **Weyl equidistribution**: generate $\varphi_0^{(n)} = (r\,n \bmod 1)\cdot\pi - \pi/2$ with the relative-phase ratio $r = (\sqrt{5}-1)/2$ (irrational) and confirm the star-discrepancy converges to 0 as $N_{\text{beats}} \to \infty$ (§5 table). For rational r it does not converge (counter-example).
2. **Position-resolved intensity**: normalize $|(\frac{\pi}{2})\psi_{\text{base}}(\varphi_0)|^2$ at each φ_0 and confirm machine-precision agreement with $|\psi_{\text{base}}|^2$.
3. **Single complex mode**: confirm that the squared readout is position-uniform ($=1$) for $\psi = e^{ik\varphi}$.

These are reproduced by the verification code as machine precision / classical theorems. The bridge (intensity → single-shot probability) is outside numerical verification and can be questioned only conceptually and experimentally (the finite-N correction).

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(This is a conjecture paper derived from peer-review dialogue notes (Noriaki Kihara × Iris, 2026-06-27). It consists of a (conditional on O1) proven core (the reproducing-kernel property + the position-resolved intensity $= |\psi_{\text{base}}|^2$ *via Weyl equidistribution) and an explicitly demarcated unproven bridge (intensity* $\to$ *single-click probability), the latter connecting to lineage A (Khrennikov-type threshold detection). B1, the exponent 2, and the scale attribution are left unsolved.)*